Individual Reflection – Chao Chen

1. Challenges with Permissions and Registries

During the testing phase, I encountered a significant technical hurdle regarding access control. My initial plan was to push the Docker image directly to the Google Artifact Registry used by my teammate, Alex. However, due to strict permission settings within the Google Cloud Platform (GCP) project, I was unable to complete the push to the original private registry.
To solve this, I had to pivot my strategy. I pulled the image locally and then pushed it to **Docker Hub**, which is a public registry. This allowed me to successfully bypass the permission issues and ensured the image was accessible for my own deployment script. This taught me that while Docker makes applications portable, managing the "gatekeepers" (permissions and registries) is a critical part of the process.

2. The Complexity of the Container Ecosystem

Working on this project helped me realize that Docker is only one piece of a much larger puzzle. Even though Docker allows a machine learning pipeline to run across different systems, it cannot function in a vacuum. It requires a whole set of supporting infrastructure, such as image repositories (like Docker Hub or Artifact Registry) and container orchestration tools like Kubernetes (k8s) or Cloud Run to manage the life cycle of the container.
While these tools provide great scalability and reliability, they also significantly increase the complexity of the project and the learning curve for a developer. Transitioning from a simple Python script to a fully Dockerized cloud application requires a deep understanding of networking, environment variables, and cloud-native services.